

# AMD AI DEVMASTER HACKATHON 2026



## StoryCanvas AI

An AI-Powered Multimodal Story Creation Platform  
Optimized for AMD Radeon GPUs

### Track 1: Multimodal Content Creation

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Date

**05 / 08 / 2026**

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# 1. Executive Summary

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StoryCanvas AI is an AI-powered multimodal content creation platform that transforms a simple natural language prompt into a fully illustrated and narrated digital storybook. The project demonstrates how multiple state-of-the-art artificial intelligence models can be orchestrated within a unified pipeline to automate the entire storytelling process, eliminating the need for separate tools for writing, illustration, narration, and publishing.

Traditional digital story creation requires significant manual effort, including creative writing, graphic design, voice recording, and document formatting. This process is often time-consuming, expensive, and inaccessible to educators, parents, students, independent authors, and content creators who may lack expertise in one or more of these areas. StoryCanvas AI addresses these challenges by providing a single platform capable of generating complete multimedia storybooks from a single user prompt.

The system combines three specialized AI models, each responsible for a specific stage of content generation:

- **Qwen2.5** generates a structured, coherent, and contextually rich multi-page story from a natural language prompt.
- **FLUX.1-schnell** creates high-quality illustrations for every page, ensuring that each image aligns with the corresponding story content.
- **Kokoro Text-to-Speech (TTS)** converts each page of the story into natural-sounding narration, making the final storybook more engaging and accessible.

These models are integrated into a modular pipeline managed by a FastAPI backend. Once a user submits a story prompt through the React-based web interface, the backend initiates an asynchronous background job that coordinates each stage of the generation process. Instead of requiring the user to wait for a long-running synchronous request, the application immediately returns a job identifier and continuously reports progress through a dedicated status endpoint. This asynchronous architecture improves responsiveness, prevents request timeouts, and provides a smoother user experience during computationally intensive AI inference.

Throughout the generation pipeline, the application maintains a structured **Story JSON** object that acts as the central data representation for the entire workflow. This object stores the generated story text, page information, image references, audio references, and metadata, enabling seamless communication between different modules such as image generation, narration, PDF generation, video creation, and the frontend story viewer.

Upon completion of the generation process, users receive an interactive digital storybook through the React frontend, where they can read the generated story, view AI-created illustrations, and listen to synchronized narration for each page. Additionally, StoryCanvas AI automatically generates two downloadable formats:

- A professionally formatted **PDF storybook** for printing or offline reading.
- A narrated **MP4 video** that combines illustrations with synthesized speech to create an engaging multimedia presentation suitable for sharing across digital platforms.

A key objective of StoryCanvas AI is to demonstrate efficient multimodal AI inference on **AMD Radeon GPUs**. The system is designed using **ROCm-compatible PyTorch**, enabling GPU-accelerated inference for computationally intensive tasks such as story generation, image synthesis, and speech generation. To maximize hardware utilization, the application employs persistent model loading, ensuring that AI models remain resident in GPU memory instead of being reloaded for every request. Combined with FP16 (half-precision) inference where supported, these optimizations reduce model initialization overhead, improve throughput, and lower GPU memory consumption, resulting in faster and more efficient content generation.

The modular architecture of StoryCanvas AI allows each AI component to operate independently while remaining part of a unified pipeline. This design simplifies future enhancements, including multilingual story generation, character consistency across illustrations, animated storytelling, collaborative editing, cloud deployment, and integration with additional multimodal AI models.

StoryCanvas AI demonstrates the practical integration of large language models, generative image models, and speech synthesis technologies into a single end-to-end application. By combining modern AI capabilities with an asynchronous, scalable software architecture optimized for AMD Radeon GPUs, the project showcases how multimodal artificial intelligence can simplify creative workflows and make high-quality digital content creation accessible to a broader audience.

## 2. Problem Statement

### 2.1 Current Challenges Digital Storybook Creation :

The creation of engaging digital storybooks is a multidisciplinary process that traditionally requires expertise in creative writing, graphic design, audio production, and digital publishing. Whether the goal is to produce educational content, children's storybooks, interactive learning materials, or multimedia presentations, creators are often required to use multiple independent software tools to complete a single project.

A typical story creation workflow consists of several distinct stages:

- Writing and structuring the story
- Designing or generating illustrations for each scene
- Recording narration or voice-overs
- Formatting the content into a printable or digital layout
- Producing a shareable document, presentation, or video

Each of these tasks generally relies on different applications, workflows, and skill sets. As a result, creators spend a significant amount of time switching between tools, manually transferring content, and ensuring consistency across text, images, and audio. This fragmented workflow increases development time and often limits creativity, especially for users without professional design or multimedia experience.

Several challenges arise from this traditional approach:

- **Time-consuming workflow:** Producing a complete storybook often requires hours or even days of work because writing, illustration, narration, and publishing are performed separately.
- **Requirement for multiple creative skills:** Developing a high-quality multimedia story demands expertise in storytelling, digital art, audio recording, and document design, making the process difficult for individuals with limited technical or artistic backgrounds.
- **High production cost:** Hiring professional writers, illustrators, and voice artists significantly increases the cost of producing educational or entertainment content, making large-scale story creation economically challenging.
- **Limited accessibility:** Teachers, parents, students, independent authors, and small content creators frequently lack access to the specialized software, computational resources, or creative expertise needed to produce professional-quality digital storybooks.
- **Inconsistent multimedia content:** When text, illustrations, and narration are created independently using different tools, maintaining consistency between scenes, characters, and storytelling style becomes difficult. This often results in a fragmented user experience.
- **Complex publishing process:** After generating individual assets, users must manually assemble them into a coherent storybook or video presentation, requiring additional editing software and technical knowledge.

### 2.2 Limitations of Current AI Tools :

| AI Tool                           | Limitation                                                                                                                  |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Large Language Models (LLMs)      | Generate high-quality stories but cannot automatically create matching illustrations or narration.                          |
| image Generation Models           | Produce high-quality images but require users to manually design prompts for every scene and integrate them with the story. |
| Text-to-Speech Systems            | Generate natural narration but depend entirely on externally generated textual content.                                     |
| Document & Video Editing Software | Require manual assembly of stories, illustrations, and narration into a final PDF or video.                                 |

## 3. Target Users & Application Scenarios

### 3.1 Target Users :

StoryCanvas AI is designed to simplify multimedia content creation for users who require high-quality illustrated and narrated storybooks without extensive expertise in creative writing, digital illustration, or audio production. By automating the complete storytelling workflow, the platform enables users from diverse domains to transform simple natural language prompts into fully illustrated multimedia stories. The primary target users are summarized below.

| Target User                     | Application                                                                                                            |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Educators</b>                | Create interactive educational stories, classroom activities, and learning materials with illustrations and narration. |
| <b>Parents &amp; Guardians</b>  | Generate personalized bedtime stories and engaging reading experiences tailored to children's interests.               |
| <b>Children's Authors</b>       | Rapidly prototype story concepts, visualize scenes, and create narrated storybooks before publication.                 |
| <b>Content Creators</b>         | Produce multimedia storytelling content for educational platforms, blogs, and social media.                            |
| <b>Educational Institutions</b> | Develop accessible digital learning resources that combine text, images, and speech for enhanced student engagement.   |

### 3.2 Application Scenarios :

StoryCanvas AI can be applied across multiple educational, creative, and digital publishing environments where rapid multimedia content generation is required. The application supports several practical scenarios.

| Application Scenario                | Description                                                                                                     |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Digital Education</b>            | Automatically generate illustrated learning stories to improve student engagement and classroom interaction.    |
| <b>Children's Storytelling</b>      | Create personalized storybooks with synchronized narration for young readers.                                   |
| <b>Creative Writing Assistance</b>  | Assist writers by converting initial ideas into complete illustrated story drafts.                              |
| <b>Digital Publishing</b>           | Accelerate the production of multimedia books for online publishing platforms.                                  |
| <b>Accessible Learning</b>          | Support visually impaired learners through AI-generated narration accompanying illustrated educational content. |
| <b>AI-Assisted Content Creation</b> | Enable rapid generation of multimedia assets from a single natural language prompt.                             |

StoryCanvas AI demonstrates how multimodal artificial intelligence can significantly simplify the production of rich digital storytelling content while reducing both development time and creative effort.

## 4. Proposed Solution

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### 4.1 Overview :

StoryCanvas AI is an end-to-end multimodal content creation platform designed to automate the complete digital storytelling workflow. The system integrates advanced Artificial Intelligence models into a unified pipeline capable of transforming a simple natural language prompt into a fully illustrated, narrated, and exportable digital storybook.

Unlike conventional content creation workflows that require multiple independent tools for writing, illustration, narration, and publishing, StoryCanvas AI performs all these tasks within a single application. This integrated approach minimizes manual effort, reduces production time, and enables users to generate professional-quality multimedia content with minimal technical expertise.

### 4.2 Solution Workflow :

The platform follows a structured, automated workflow consisting of the following stages:

#### 1. Accept User Prompt :

The user provides a natural language description of the desired story through the React-based web interface.

**Example:**

*"A brave little fox discovers a magical forest and helps its creatures defeat an ancient shadow monster."*

#### 2. Generate Story :

The prompt is processed by the Qwen2.5 Large Language Model, which generates:

- Story title
- Genre
- Multi-page story content
- Structured JSON representation

#### 3. Generate Illustrations :

Each story page is converted into a descriptive image prompt and the FLUX.1-schnell image generation model creates a unique illustration corresponding to each page.

#### 4. Generate Narration :

The generated story text is passed to the Kokoro Text-to-Speech (TTS) model.

Natural-sounding narration is synthesized for every page and stored as individual audio files.

#### 5. Generate Output Assets :

The generated text, illustrations, and narration are combined to produce:

A professionally formatted PDF storybook

A narrated MP4 video

A structured Story JSON file containing all generated assets and metadata.

#### 6. Display Interactive Story :

The completed story is presented through a modern React-based Story Viewer, allowing users to:

- ✓ Read the generated story
- ✓ View illustrations
- ✓ Listen to narration
- ✓ Download the generated PDF and MP4 video

### 4.3 System Design Approach :

To improve scalability and user experience, StoryCanvas AI adopts an asynchronous processing architecture.

Instead of executing all AI tasks within a single HTTP request, the backend creates a background job that independently executes the complete AI pipeline. The frontend immediately receives a unique Job ID and periodically queries the backend to retrieve the current processing status.

This architecture provides several advantages:

- ❖ Eliminates HTTP request timeout issues.
- ❖ Maintains a responsive user interface during long-running AI inference.
- ❖ Supports efficient execution of computationally intensive AI models.
- ❖ Enables progress tracking throughout the generation process.
- ❖ Simplifies future scaling to support multiple concurrent users.

### 4.4 Interface Preview :

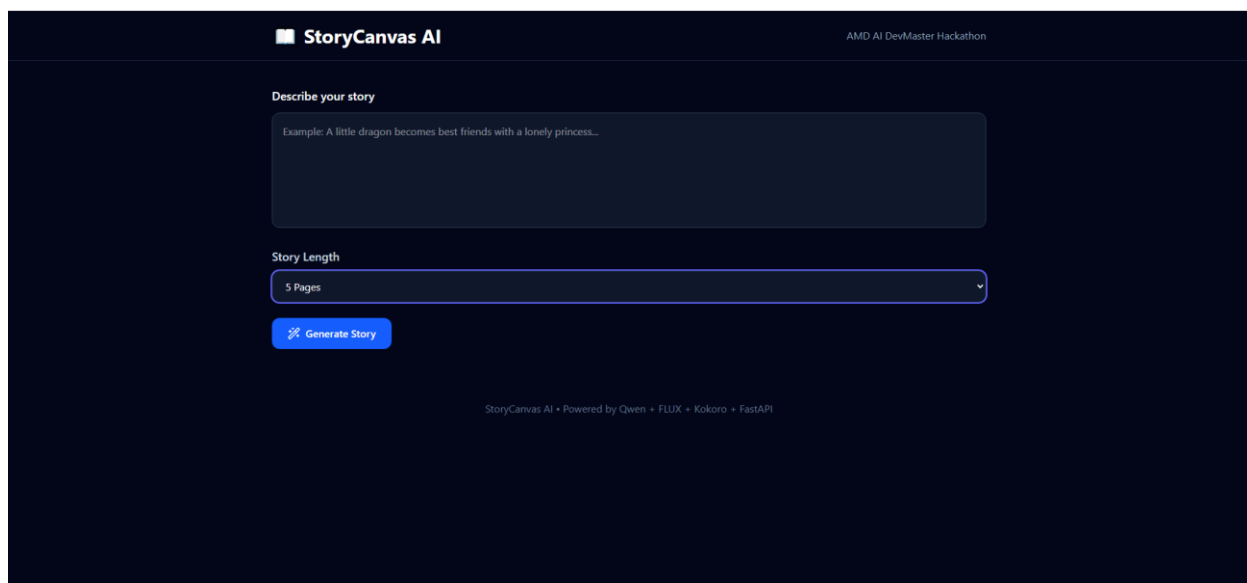


Figure 1 : Home Page

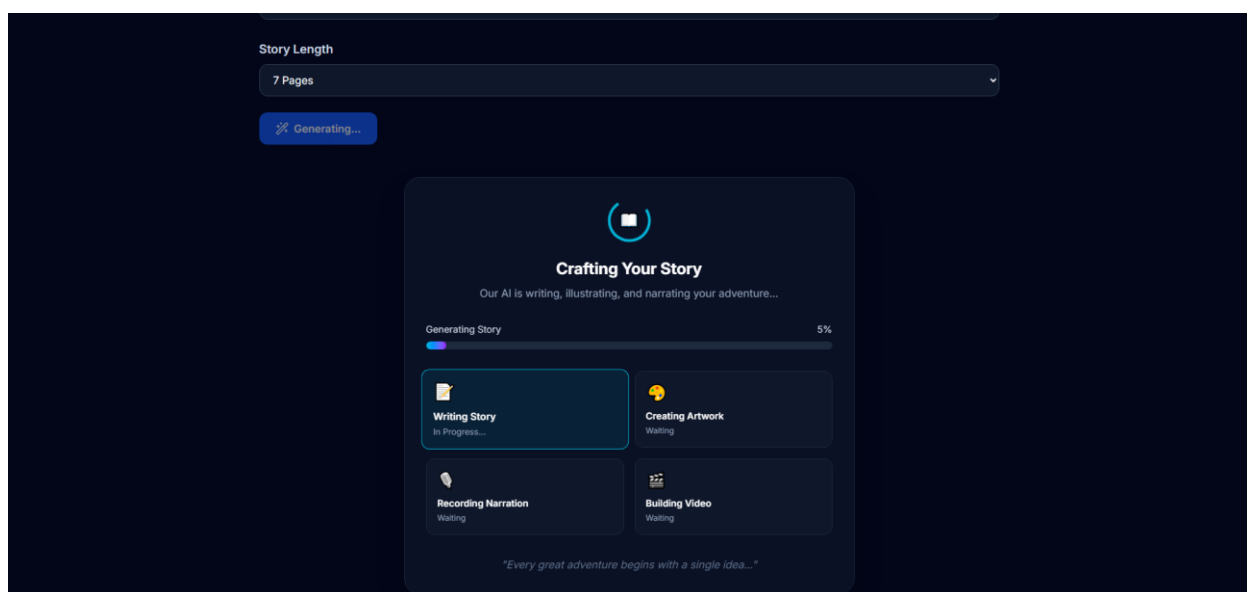


Figure 2 : Generating Screen



## 5. System Architecture

### 5.1 Architecture Overview :

StoryCanvas AI follows a modular, layered architecture that separates user interaction, backend services, AI processing, and output generation into independent components. This design improves maintainability, scalability, and extensibility while enabling computationally intensive AI tasks to execute efficiently without affecting the responsiveness of the user interface.

The architecture is organized into four primary layers:

1. Frontend Layer
2. Backend Layer
3. AI Processing Layer
4. Output Layer

Each layer performs a dedicated responsibility while communicating through well-defined interfaces. This separation of concerns allows individual modules to be modified, upgraded, or replaced without impacting the overall system.

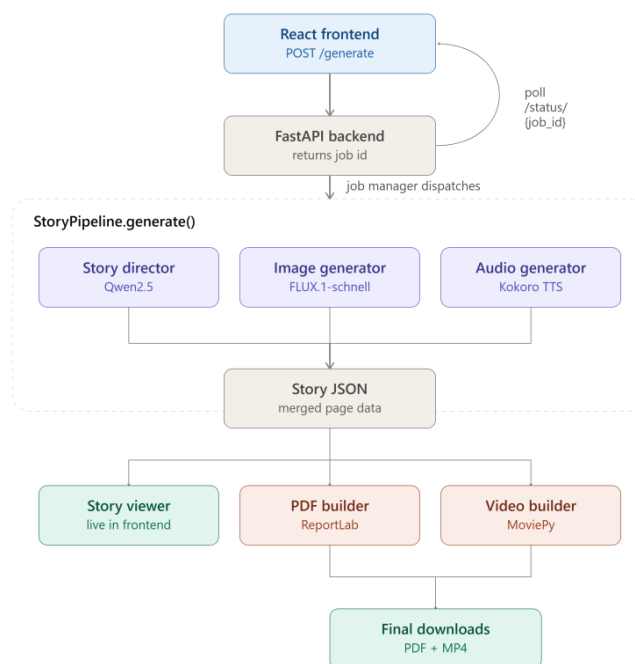


Figure 3 : System Architecture of StoryCanvas AI

### 5.2 Frontend Layer :

The frontend of StoryCanvas AI is developed using **React**, **Vite**, and **Tailwind CSS**, providing a responsive and user-friendly interface for interacting with the application. It serves as the primary interaction point between the user and the AI pipeline.

The frontend allows users to submit story prompts, monitor the generation progress, visualize the generated story with illustrations and narration, and download the generated PDF and MP4 outputs. Communication with the backend is performed through REST APIs using asynchronous HTTP requests. To avoid blocking the user interface during AI inference, the frontend follows an asynchronous workflow. After submitting a generation request, it receives a unique Job ID and periodically queries the backend for the current job status until the story generation process is completed.

### 5.3 Backend Layer :

The backend is implemented using **FastAPI** and acts as the central controller of the application. It exposes RESTful APIs for handling client requests and coordinates the execution of the complete AI pipeline.

Upon receiving a story generation request, the backend creates a background job and immediately returns a Job ID to the frontend. The generation process is then executed asynchronously by the Job Manager, which tracks the execution state and stores intermediate and final outputs. This design prevents request timeouts and enables continuous progress updates throughout the generation process.

The backend is also responsible for serving generated media files, managing the Story JSON structure, and providing downloadable PDF and video outputs.

### 5.4 AI Processing Layer :

The AI Processing Layer is responsible for executing the multimodal content generation pipeline. The entire workflow is orchestrated by the **StoryPipeline**, which sequentially invokes independent processing modules responsible for story generation, illustration generation, narration synthesis, and output creation. The **Story Director** generates a structured multi-page story using the **Qwen2.5** large language model. The generated story is then processed by the **Image Generator**, which produces context-aware illustrations using **FLUX.1-schnell**. Subsequently, the **Audio Generator** synthesizes natural narration for each page using the **Kokoro Text-to-Speech** model.

The generated text, images, and audio are consolidated into a centralized **Story JSON** object, which serves as the common data representation throughout the pipeline. Finally, the **PDF Builder** and **Video Builder** utilize this structured data to generate downloadable storybook and video outputs.

### 5.5 Output Layer :

The Output Layer assembles all generated assets into user-consumable formats. Once the AI pipeline completes execution, the generated story, illustrations, narration, and associated metadata are made available through the React Story Viewer.

In addition to the interactive web interface, the application automatically generates a printable **PDF storybook** and a narrated **MP4 video**, allowing users to access and share the generated content across different platforms. The structured Story JSON is preserved as the central representation of the generated story, enabling seamless integration between different output modules.

## 6. Software Architecture

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### 6.1 Overview :

StoryCanvas AI follows a **modular software architecture**, where each functional component is responsible for a specific stage of the multimodal content generation pipeline. The application is designed using the principle of **separation of concerns**, allowing each module to perform an independent task while communicating through a centralized data structure. This modular approach improves maintainability, simplifies debugging, and enables individual components to be extended or replaced without affecting the overall system.

The complete workflow is coordinated by the **StoryPipeline**, which orchestrates the execution of all processing modules in a predefined sequence. Each module contributes its output to a shared **Story JSON** object, ensuring consistent data exchange throughout the pipeline.

### 6.2 Story Director :

The Story Director is the entry point of the AI processing pipeline and is responsible for generating the textual content of the story. It receives the user's natural language prompt and processes it using the **Qwen2.5 Large Language Model** to generate a structured story.

Rather than producing plain text, the Story Director creates a structured representation containing the story title, genre, page-wise narrative, and additional metadata required for downstream processing. This

structured output establishes the foundation for all subsequent modules, ensuring that image generation, narration synthesis, and document generation operate on a consistent representation of the story. By separating story generation from other processing stages, the Story Director enables different language models or prompt engineering strategies to be integrated into the system without affecting the remaining architecture.

### 6.3 Image Generator :

The Image Generator is responsible for transforming the textual description of each story page into visually meaningful illustrations. It receives the structured story generated by the Story Director and extracts the narrative of each page to construct descriptive image prompts suitable for the image generation model.

These prompts are processed by **FLUX.1-schnell**, which generates a high-quality illustration for every page while preserving the context of the corresponding narrative. The generated images are stored as media assets, and their references are added to the centralized Story JSON object for use by the Story Viewer, PDF Builder, and Video Builder.

Separating image generation into an independent module allows future support for additional image generation models or different artistic styles without requiring modifications to the remaining components.

### 6.4 Audio Generator :

The Audio Generator converts the textual content of each story page into natural speech narration. It processes the page-wise story content using the **Kokoro Text-to-Speech (TTS)** model and generates individual audio files corresponding to each page.

The generated narration enhances the storytelling experience by making the application accessible to children, educators, and visually impaired users. Audio references are subsequently incorporated into the Story JSON structure, enabling synchronized playback within the Story Viewer and narrated video generation.

The modular design of the Audio Generator allows different speech synthesis models, languages, or voice profiles to be integrated with minimal changes to the overall software architecture.

### 6.5 PDF Builder :

The PDF Builder is responsible for generating a printable version of the completed storybook. It combines the story title, page-wise narrative, and corresponding illustrations into a professionally formatted PDF document.

Using the structured Story JSON as its primary data source, the PDF Builder organizes the generated content into a consistent layout suitable for offline reading, printing, and digital distribution. Since the PDF generation process is independent of the AI inference modules, improvements to document formatting or additional export templates can be introduced without affecting the content generation pipeline.

### 6.6 Video Builder :

The Video Builder creates a narrated multimedia presentation by synchronizing the generated illustrations with their corresponding narration. It combines page images and synthesized audio into a sequential MP4 video, providing an engaging visual representation of the generated story.

The module automatically determines the display duration of each page based on the narration length, ensuring synchronization between visual and audio content. This design enables StoryCanvas AI to produce shareable multimedia content suitable for educational presentations, storytelling sessions, and digital publishing platforms.

6.7 Job Manager :

The Job Manager is responsible for coordinating long-running AI inference tasks while maintaining application responsiveness. Instead of executing the complete generation pipeline within a single HTTP request, the Job Manager creates an asynchronous background task that independently executes the StoryPipeline.

Upon receiving a generation request, the backend immediately returns a unique **Job ID** to the frontend. The Job Manager continuously updates the execution status, progress information, and generated outputs associated with the job. The frontend periodically retrieves this information to provide real-time progress updates without blocking the user interface.

This asynchronous execution model eliminates request timeout issues, improves scalability, and enables efficient management of computationally intensive AI workloads. It also provides a flexible foundation for supporting multiple concurrent users in future deployments.

6.8 Software Design Summary :

The software architecture of StoryCanvas AI is designed to achieve modularity, maintainability, and scalability. Each processing module performs a well-defined responsibility while interacting through a shared Story JSON data structure managed by the StoryPipeline. This architecture minimizes coupling between components, simplifies future enhancements, and enables seamless integration of additional AI models or output formats. By combining modular software design with asynchronous execution, StoryCanvas AI provides a robust and extensible framework for multimodal content creation.

7. AI Models

7.1 Overview :

StoryCanvas AI integrates three specialized Artificial Intelligence models to perform different stages of the multimodal content generation pipeline. Instead of relying on a single general-purpose model, the system adopts a modular AI architecture in which each model is optimized for a specific task. This design improves overall output quality while maintaining efficient GPU resource utilization.

Each model was selected after considering multiple factors, including inference speed, output quality, memory requirements, ease of integration, and compatibility with AMD Radeon GPUs through the ROCm ecosystem. Together, these models enable StoryCanvas AI to generate coherent stories, context-aware illustrations, and natural speech narration within a unified pipeline.

7.2 AI Model Selection :

| Model          | Primary Function           | Role in StoryCanvas AI                                                                                                          | Reason for Selection                                                                                                                                                                                          |
|----------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Qwen2.5        | Large Language Model (LLM) | Generates the story title, genre, and structured multi-page narrative from the user's prompt.                                   | Chosen for its excellent instruction-following capability, coherent long-form text generation, strong contextual understanding, and ability to produce structured outputs suitable for downstream AI modules. |
| FLUX.1-schnell | Text-to-Image Generation   | Generates high-quality illustrations for each story page based on the generated narrative.                                      | Selected for its ability to produce visually detailed images with fewer inference steps, providing a good balance between image quality, generation speed, and GPU efficiency for interactive applications.   |
| Kokoro TTS     | Text-to-Speech Synthesis   | Converts each page of the generated story into natural-sounding narration used in both the Story Viewer and narrated MP4 video. | Chosen because of its lightweight architecture, low inference overhead, fast synthesis speed, and clear, natural speech quality suitable for continuous page-wise narration.                                  |

### 7.3 Model Integration within the Pipeline :

Rather than operating independently, the three AI models are integrated into a sequential processing pipeline coordinated by the **StoryPipeline** module.

The workflow begins with **Qwen2.5**, which interprets the user's prompt and generates a structured story consisting of a title, genre, and multiple story pages. The generated narrative then serves as the input for **FLUX.1-schnell**, which creates page-specific illustrations by converting each page into an optimized image prompt. Finally, the same story content is processed by **Kokoro TTS** to synthesize natural narration for every page.

The outputs produced by each model are consolidated into a centralized **Story JSON** object, enabling seamless communication between the image generation, narration, PDF generation, video generation, and frontend rendering modules.

### 7.4 Model Selection Considerations :

The selection of AI models was guided by both functional and system-level requirements. Since all models execute within a shared inference pipeline, it was essential to balance generation quality with computational efficiency.

The following design considerations influenced the model selection:

| Selection Criterion    | Design Consideration                                                                                                         |
|------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Output Quality         | Generate coherent stories, visually consistent illustrations, and natural speech suitable for digital storytelling.          |
| Inference Performance  | Maintain low latency to provide an interactive user experience during story generation.                                      |
| GPU Memory Utilization | Ensure efficient execution of multiple AI models within the available GPU memory.                                            |
| Pipeline Compatibility | Produce structured outputs that can be seamlessly consumed by downstream modules without additional preprocessing.           |
| Scalability            | Support future replacement or upgrading of individual models without requiring changes to the overall software architecture. |
| ROCm Compatibility     | Enable GPU-accelerated inference using AMD Radeon GPUs through ROCm-compatible PyTorch.                                      |

### 7.5 Summary :

The combination of **Qwen2.5**, **FLUX.1-schnell**, and **Kokoro TTS** provides a balanced multimodal AI pipeline capable of generating complete digital storybooks from a single text prompt. By assigning a specialized responsibility to each model, StoryCanvas AI achieves high-quality content generation while maintaining efficient GPU utilization and a modular software architecture. This design enables the application to deliver an end-to-end storytelling experience that integrates text generation, image synthesis, and speech generation within a single, scalable framework.

Each model was selected to balance output quality against inference cost, since all three run within the same request pipeline on shared GPU resources.

## 8. AMD GPU Adaptation

### 8.1 Overview :

StoryCanvas AI is developed and deployed on the **AMD Radeon Cloud Platform**, utilizing the **ROCm (Radeon Open Compute)** software stack to accelerate multimodal AI inference. The complete application, including story generation, image synthesis, speech generation, and media processing, executes within the ROCm-enabled cloud environment provided by AMD.

Unlike traditional CPU-based deployments, StoryCanvas AI leverages AMD Radeon GPUs as the primary compute resource for neural network inference. The application integrates ROCm-compatible AI frameworks and optimized inference pipelines to efficiently execute multiple deep learning models within a unified workflow.

The deployment architecture demonstrates the practical use of AMD's AI ecosystem for building scalable multimodal content creation applications.

### 8.2 AMD Radeon Cloud Deployment and Techstack:

| Component               | Implementation             | Purpose                                                                                 |
|-------------------------|----------------------------|-----------------------------------------------------------------------------------------|
| Deployment Platform     | AMD Radeon Cloud           | Provides a cloud-based GPU environment for executing multimodal AI workloads.           |
| GPU Computing Framework | ROCm (Radeon Open Compute) | Enables GPU-accelerated execution of deep learning models on AMD Radeon hardware.       |
| Deep Learning Framework | ROCm-Compatible PyTorch    | Executes neural network inference using AMD GPU acceleration.                           |
| Application Server      | FastAPI                    | Hosts REST APIs and manages communication between the frontend and AI pipeline.         |
| Frontend Deployment     | React + Vite               | Provides the web interface for user interaction and visualization of generated content. |

| Component     | Technology                |
|---------------|---------------------------|
| Frontend      | React, Vite, Tailwind CSS |
| Backend       | FastAPI                   |
| Story Model   | Qwen2.5                   |
| Image Model   | FLUX.1-schnell            |
| TTS           | Kokoro                    |
| PDF           | ReportLab                 |
| Video         | MoviePy                   |
| GPU Framework | ROCm                      |
| Deep Learning | PyTorch 2.9.1             |
| Deployment    | AMD Radeon Cloud          |

### 8.3 ROCm-Based AI Execution :

StoryCanvas AI is designed to execute all computationally intensive AI workloads using the ROCm software stack. The application utilizes ROCm-compatible PyTorch to offload neural network inference from the CPU to the AMD Radeon GPU, significantly improving execution efficiency for multimodal content generation.

The **Qwen2.5**, **FLUX.1-schnell**, and **Kokoro TTS** models are initialized within the ROCm runtime and execute directly on the GPU throughout the content generation pipeline. This approach reduces inference latency while allowing multiple AI models to share the same GPU resources efficiently. By adopting ROCm as the primary execution framework, the application demonstrates compatibility with AMD's open GPU computing ecosystem while maintaining portability across supported Radeon hardware.

## 8.4 GPU Optimization Techniques :

| Optimization Technique                 | Implementation                                                                                                                          | Benefit                                                                                       |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>ROCm GPU Acceleration</b>           | AI models execute through ROCm-enabled PyTorch on AMD Radeon GPUs.                                                                      | Accelerates neural network inference and reduces overall story generation time.               |
| <b>Persistent Model Loading</b>        | AI models are loaded into GPU memory during application startup and reused for all subsequent requests.                                 | Eliminates repeated model initialization overhead and improves response time.                 |
| <b>Half-Precision (FP16) Inference</b> | Supported models execute using FP16 precision during inference.                                                                         | Reduces GPU memory consumption while increasing inference throughput.                         |
| <b>Asynchronous Pipeline Execution</b> | AI inference is executed through FastAPI background jobs managed by the Job Manager.                                                    | Maintains API responsiveness and prevents HTTP request timeouts during long-running AI tasks. |
| <b>GPU-Centric AI Pipeline</b>         | Story generation, image synthesis, and speech synthesis execute on the GPU, while request handling and file management remain CPU-side. | Maximizes GPU utilization while efficiently balancing CPU and GPU workloads.                  |

## 8.5 Performance & Result:

| Metric                                  | Measured Value      |
|-----------------------------------------|---------------------|
| PyTorch Version                         | 2.9.1+ gitff65f5b   |
| GPU Platform                            | AMD Radeon Graphics |
| Available VRAM                          | 47.98 GB            |
| Qwen2.5 Loading Time (FP16)             | 25 s                |
| FLUX.1-schnell Loading Time (FP16)      | 68 s                |
| Kokoro TTS Loading Time                 | 4 s                 |
| Story Generation Time                   | 3 s                 |
| Image Generation Time                   | 3 s / page          |
| Audio Generation Time                   | 4 s / page          |
| Video Generation Time                   | 62 s                |
| Total Pipeline Execution Time           | 72 s                |
| GPU Memory Utilization – Qwen2.5        | 7.29 GB             |
| GPU Memory Utilization – FLUX.1-schnell | 33.53 GB            |
| GPU Memory Utilization – Kokoro TTS     | 0.86 GB             |
| Total GPU Memory Utilized               | 41.68 GB            |

## 9. Practical Impact & Social Value

### 9.1 Practical Impact :

StoryCanvas AI has been developed to address the practical challenges associated with multimedia content creation by integrating multiple artificial intelligence technologies into a single automated workflow. Conventional digital storybook creation typically requires separate tools for creative writing, digital illustration, voice recording, document formatting, and video production. This fragmented workflow demands considerable time, technical expertise, and financial resources, making high-quality multimedia storytelling inaccessible to many individuals and organizations.

By consolidating these independent tasks into a unified AI-driven pipeline, StoryCanvas AI significantly simplifies the content creation process. A user is only required to provide a natural language prompt describing the desired story. The system automatically generates a coherent multi-page narrative, produces context-aware illustrations for each page, synthesizes natural speech narration, and compiles the generated assets into a printable PDF storybook and a narrated MP4 video. This automation substantially reduces manual effort while maintaining a structured and intuitive user experience.

The practical applications of StoryCanvas AI extend across several domains, including education, digital publishing, creative media, and accessible learning. The platform enables users with limited artistic or technical expertise to create multimedia stories that would traditionally require collaboration among writers, illustrators, voice artists, and multimedia editors.

The practical benefits offered by the platform are summarized below.

| Application Domain              | Practical Impact                                                                                                                                                                                                   |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Education</b>                | Enables educators to rapidly develop illustrated learning materials, classroom storytelling activities, and interactive educational content without requiring professional design or multimedia production skills. |
| <b>Early Childhood Learning</b> | Supports parents and caregivers by generating personalized bedtime stories and engaging reading experiences tailored to children's interests and imagination.                                                      |
| <b>Digital Publishing</b>       | Assists authors and publishers in rapidly prototyping illustrated storybooks, reducing the time required for concept visualization and content development.                                                        |
| <b>Content Creation</b>         | Simplifies multimedia production for bloggers, educators, social media creators, and digital storytellers by automatically generating complete visual and narrated content.                                        |
| <b>Accessibility</b>            | Improves accessibility by combining written stories with AI-generated speech narration, supporting learners and readers who benefit from audio-assisted content.                                                   |

The integration of these capabilities into a single platform demonstrates how multimodal artificial intelligence can streamline creative workflows while making high-quality digital storytelling more efficient, affordable, and widely accessible.

### 9.2 Social Value :

StoryCanvas AI contributes to making creative content generation more accessible by lowering the technical and artistic barriers traditionally associated with multimedia storytelling. Individuals without professional writing, illustration, or audio production skills can create high-quality digital storybooks using natural language prompts.

The platform encourages creativity, supports inclusive learning environments, and demonstrates how generative artificial intelligence can be applied responsibly to enhance education, digital literacy, and creative expression.



## 10. Future Scope & Conclusion

### 10.1 Future Optimization Opportunities :

While the current implementation prioritizes complete multimodal functionality and high-quality content generation, additional optimization techniques can further improve deployment efficiency and resource utilization. Future work will investigate lightweight inference strategies, including model quantization, memory-efficient attention mechanisms, and optimized image generation pipelines to reduce GPU memory consumption while preserving output quality.

Furthermore, advances in reference-guided image generation and persistent character embeddings can improve visual consistency across story pages, enabling stable character appearance throughout long-form narratives. These optimizations will strengthen the scalability of StoryCanvas AI while supporting deployment on a wider range of AMD Radeon GPU configurations.

### 10.2 Future Scope :

StoryCanvas AI has been designed with a modular and extensible architecture, allowing new capabilities to be integrated with minimal modifications to the existing system. The current implementation establishes a strong foundation for future research and development in AI-powered multimedia content generation.

Several enhancements can further improve the functionality and usability of the application.

| Enhancement Area                     | Description                                                                                                                                                             |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Multilingual Story Generation</b> | Extend the language model pipeline to generate complete stories in multiple regional and international languages, making the platform accessible to a broader audience. |
| <b>Interactive Story Editing</b>     | Enable users to modify story content, regenerate individual pages, or customize prompts without recreating the entire story, providing greater creative control.        |
| <b>Character Consistency</b>         | Incorporate character-aware image generation techniques to maintain consistent appearance, clothing, expressions, and visual identity across all story illustrations.   |
| <b>Animation Generation</b>          | Extend static illustrations into short animated sequences using AI-based image-to-video models, creating richer and more engaging storytelling experiences.             |
| <b>Cloud-Scale Deployment</b>        | Deploy the application using containerized microservices with GPU orchestration to support concurrent users, improved scalability, and production-ready performance.    |
| <b>Collaborative Storytelling</b>    | Introduce multi-user collaboration, allowing authors, educators, and creative teams to jointly create, edit, and review stories in real time.                           |

The modular software architecture adopted in StoryCanvas AI enables these enhancements to be incorporated independently without requiring significant redesign of the existing pipeline.

### 10.3 Conclusion :

StoryCanvas AI demonstrates the successful integration of **Large Language Models (LLMs)**, **Text-to-Image Generation**, and **Neural Text-to-Speech** technologies into a unified multimodal content creation system. By combining these AI capabilities within a structured processing pipeline, the application transforms a simple natural language prompt into a complete digital storybook containing coherent narrative text, AI-generated illustrations, synchronized speech narration, printable PDF documents, and narrated MP4 videos.

The system adopts a modular software architecture coordinated by an asynchronous processing pipeline, ensuring clear separation of responsibilities between story generation, image synthesis, narration, and output generation. This design improves maintainability, scalability, and extensibility while providing a responsive user experience through background job execution and real-time progress monitoring.

Furthermore, the application leverages the **AMD Radeon Cloud Platform** and the **ROCm software ecosystem** to accelerate AI inference using GPU-enabled PyTorch. Techniques such as persistent model loading, half-precision inference, and efficient CPU-GPU workload distribution contribute to improved execution efficiency and demonstrate practical utilization of AMD GPU resources for multimodal AI applications.

Overall, StoryCanvas AI showcases how modern generative AI technologies can be integrated into a scalable, end-to-end storytelling platform. The project not only addresses the challenge of automated multimedia content creation but also establishes a flexible foundation for future advancements in AI-assisted education, digital publishing, creative content generation, and interactive storytelling.